TWITTER STOCK MARKET ANALYSIS REPORT

Introduction:

This project analyses Twitter's historical stock market performance using Python. The objective is to study price trends, volatility, trading volume, and investor behaviour from listing to delisting.

Methodology:

The project was done doing the following steps:

1.Data Collection: Stock market data of Twitter was collected, information regarding opening price, closing price, high low, adjusted close and volume.

2.Data preprocessing: The dataset was examined for missing values and inconsistencies. Rows containing null values were removed to ensure data accuracy. The date column was converted into a datetime format and the dataset was sorted chronologically to support times-series analysis.

3.Statistical Analysis: Statistical tests such as the t-test were applied to analyze difference between High, Low and Close process. The Chi-Square test was don't to determine the dependency between stock prices and trading volume.

4.Visualization and Trend Analysis: Line graphs were created to visualise stock price over time. Comparative plots were used to study price movements across different time periods and to identify volatility patterns.

5. Dashboard Creation: Multiple visualizations were combined into a single dashboard to provide a consolidated view of stock performance, trading volume, and price fluctuations for effective interpretation.

Requirement Analysis:

The following tools were required during the project which are as follows:

1.Data Requirements:

- Date
- Open Price
- High price
- Low Price
- Close Price
- Adjusted Close Price
- Trading Volume

2.Software Requirements:

- Python programming Language
- Google Colab

3.Python Libraries used:

- Pandas
- Matplotlib
- SciPy
- WordCloud

4.Hardware Requirements:

- Minimum 4GB RAM

Other Parameters:

1.Scope: This project focuses on analysing Twitter's stock market data using statistical and visualization techniques. The scope includes studying price movements, trading volume, volatility patterns and investor behaviour over time. The data doesn't include real time data.

2.Assumption:

- ✓ The dataset obtained is accurate and reliable and it reflects genuine market conditions.
- ✓ Statistical tests are applied under standard assumptions of independence and normal distribution where required.
- ✓ Market behavior is influenced primarily by observable stock data such as price and volume.

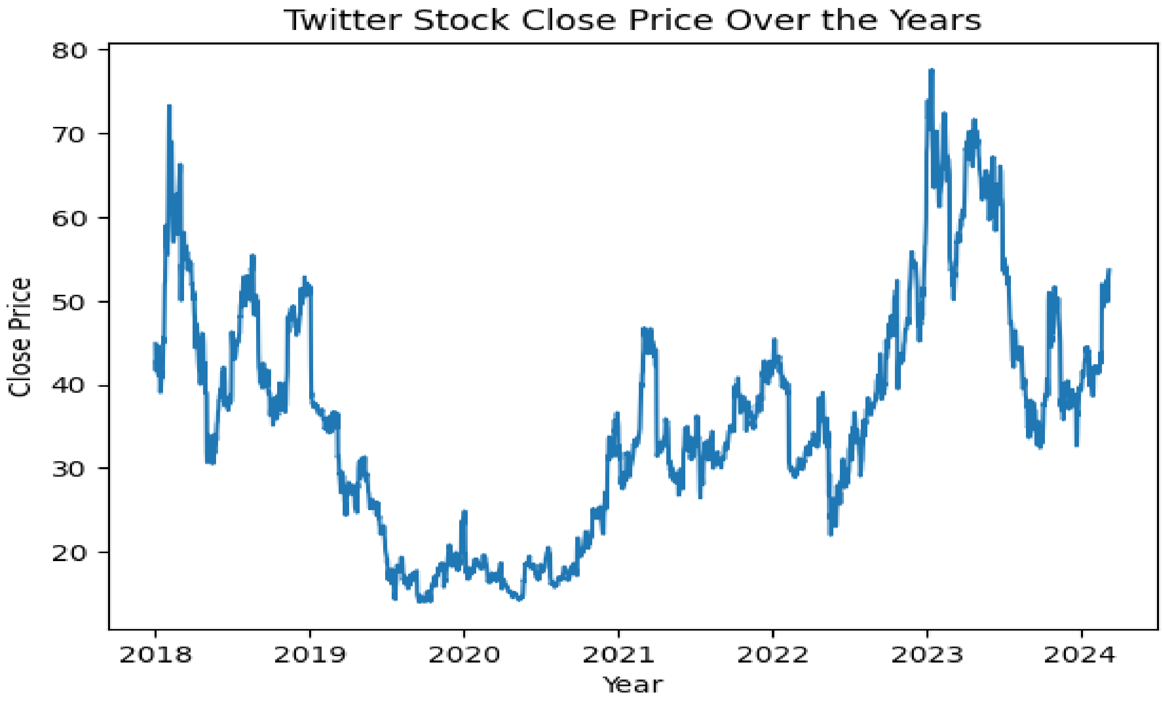
3.Limitations:

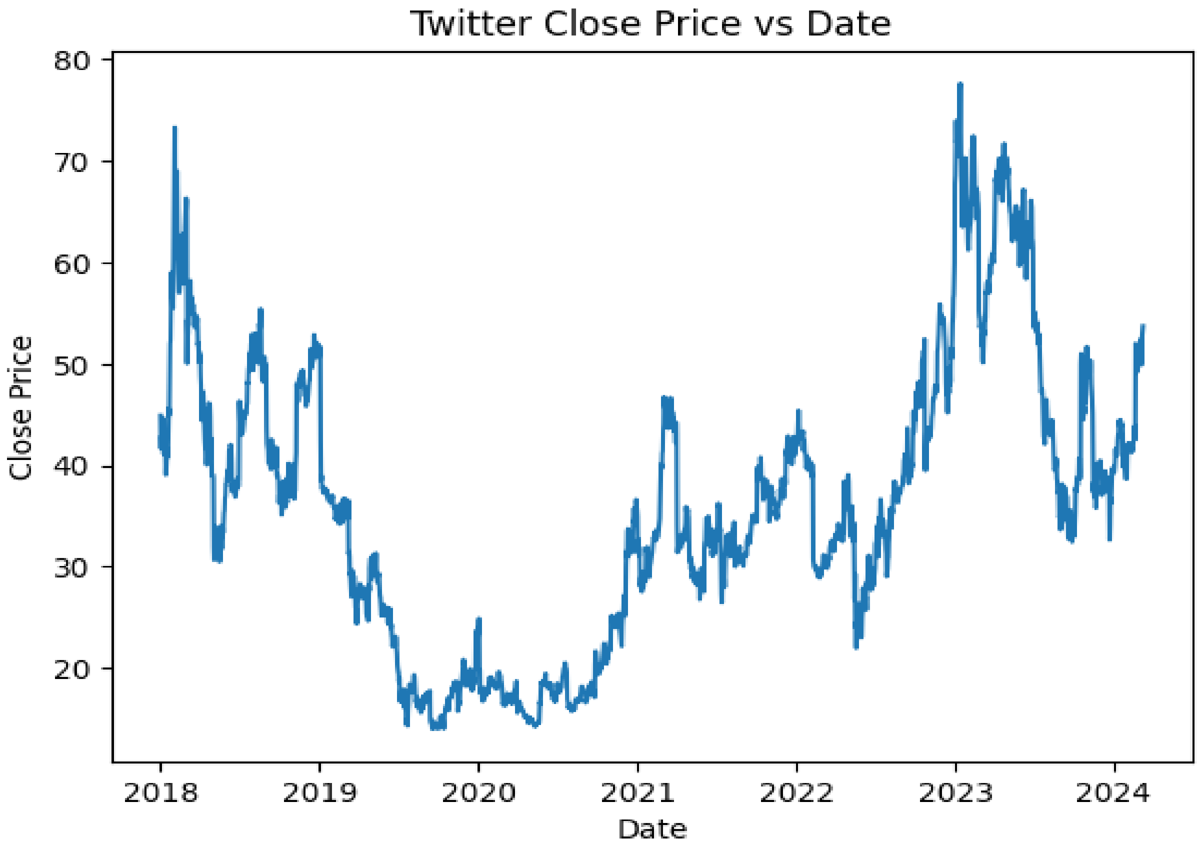
- ✓ The analysis is based only on historical data.
- ✓ External factors such as global economic events, political influences, or company internal decisions are not quantitatively modelled.
- ✓ No predictive machine learning models were used.
- ✓ The analysis does not guarantee future stock performance.

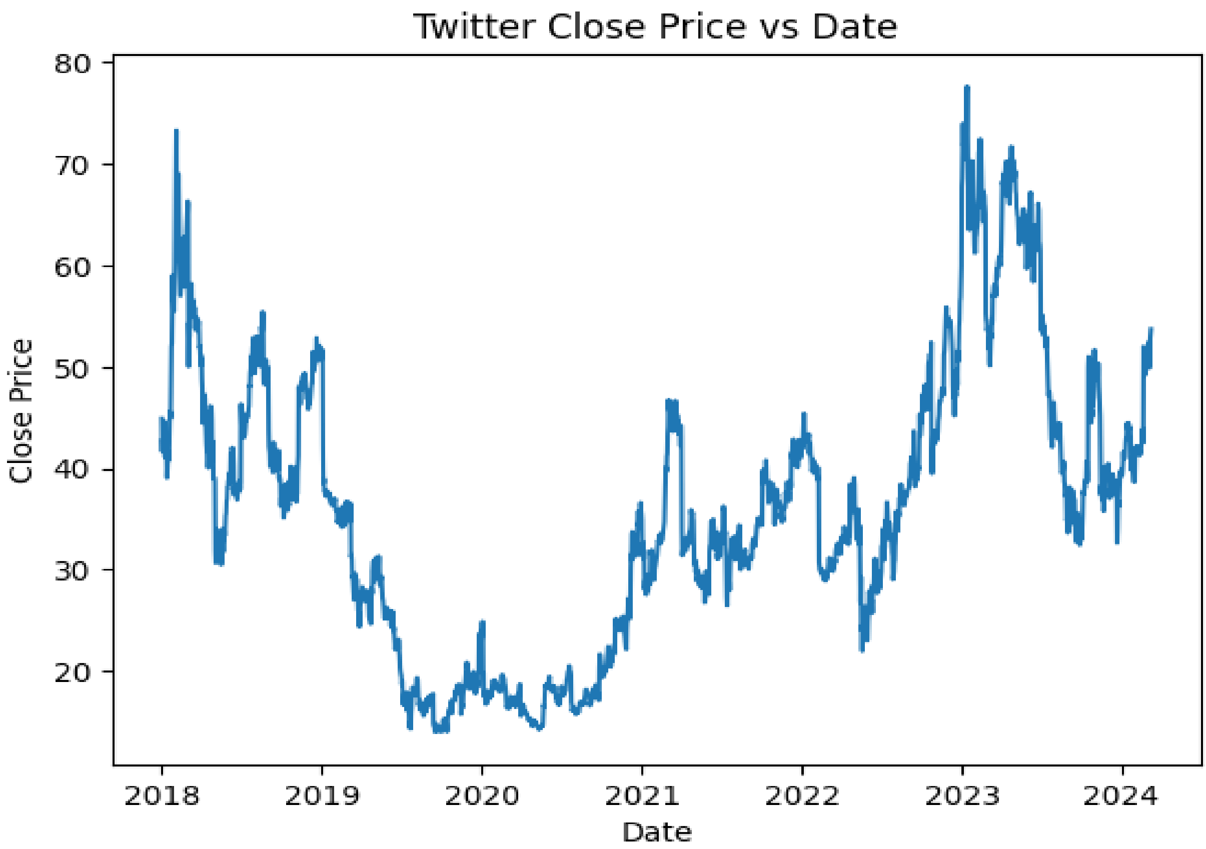
4.Constraints:

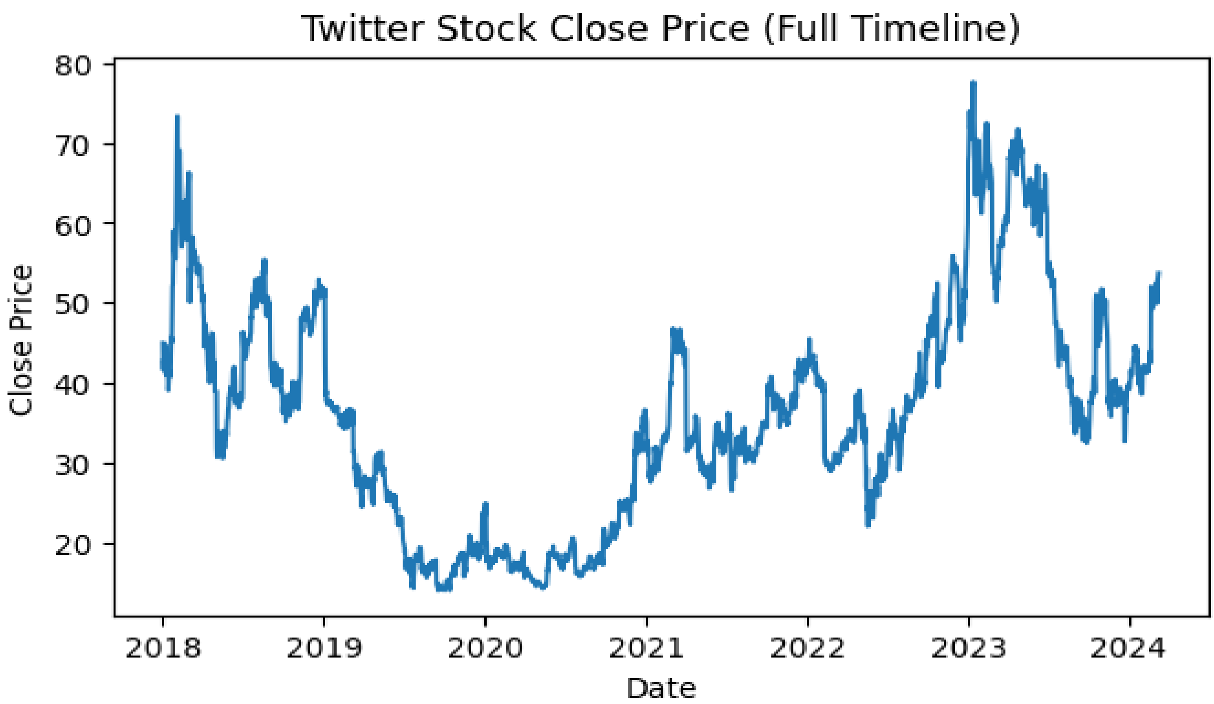
- ✓ Limited to only a specific data.
- ✓ Dependent on historical data quality.

Visualisation:

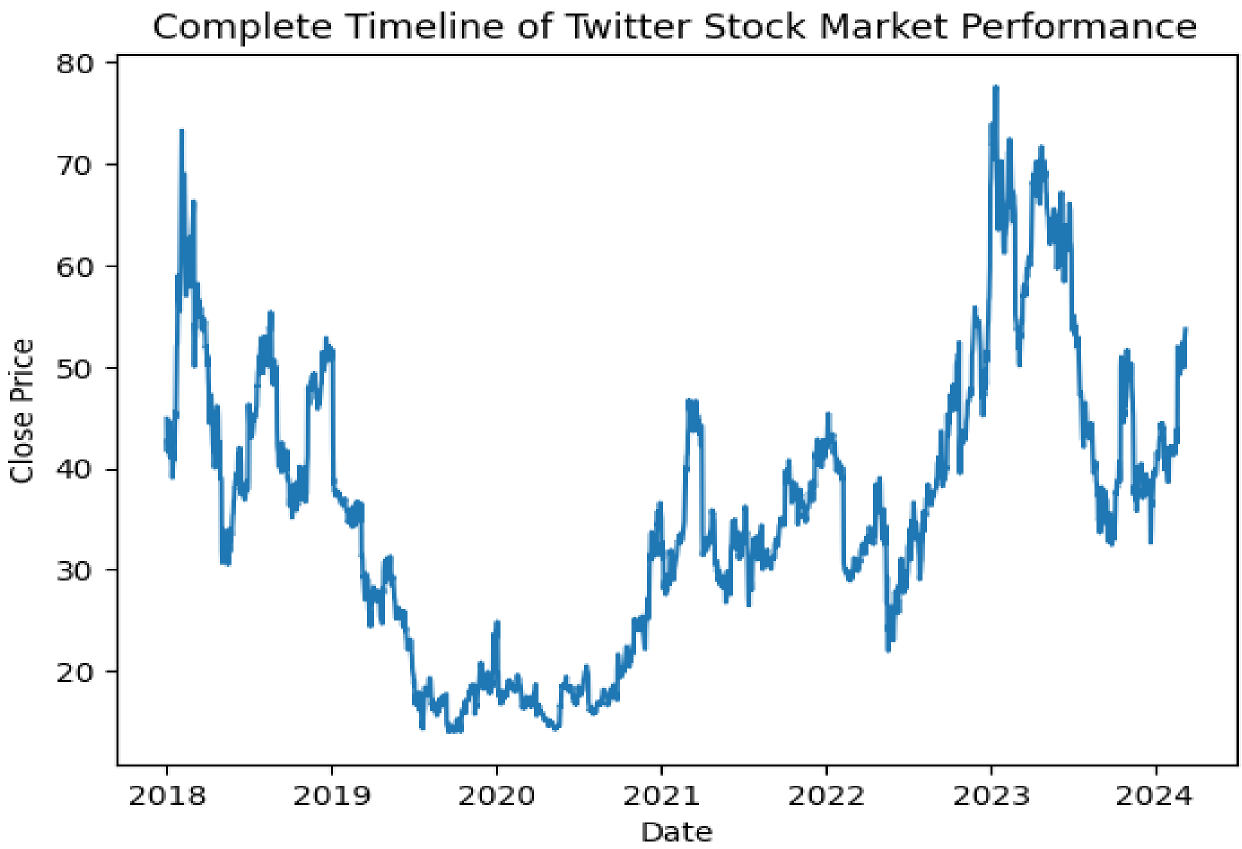








Full Period | 2018–2020 | 2021–2023


Complete Timeline of Twitter Stock Market Performance
Close Price
Date
80
70
60
50
40
30
20
2018
2019
2020
2021
2022
2023
2024


Word Cloud of Twitter Stock Market Analysis
decline
growth
term
sentiment
investor
delisting
volume
risk
uncertainty
fluctuation
instability
price
trading
long
short
volatility
market

Insights:

The visualisation clearly show frequent fluctuations in twitter's stock prices, indicating high volatility. The difference between high and low process across multiple fays reflects unstable market behaviour and sensitivity.

Although the price increases during some period of time, the overall trend does not show steady long term growth. The stock experienced repeated rises followed by declines, which tells inconsistent performance over time.

The volume analysis indicates that significant price movements often coincide with spikes in trading volume. This tells that the investor activity played a major role in influencing stock prices.

Sudden increases or drops in stock prices suggest that the market reacted strongly to external events such as company decisions.

Time-period analysis shows that recent years exhibited more intense fluctuations compared to earlier periods.

The High vs Low price comparison shows that during stable periods, the gap between high and low prices was smaller, whereas during volatile periods, the gap widened significantly.

It is evident that Twitter's stock was influenced by uncertainty, irregular growth patterns, and fluctuating investor confidence.

Conclusion:

This project analysed Twitter's historical stock market performance using data analysis, statistical techniques, and visualizations. The study revealed that Twitter's stock exhibited significant volatility, with frequent fluctuations in price over time. Despite periods of growth, the stock failed to maintain consistent long-term upward momentum.

The analysis also showed that trading volume had a strong influence on price movements. Additionally, the stock appeared to be highly sensitive to external events, leading to sudden rises and declines.

Overall, the findings suggest that Twitter's stock performance was unstable and influenced by market uncertainty and changing investor sentiment. This lack of consistency and predictability contributed to its eventual delisting from the stock exchange.

The project demonstrates how data-driven analysis can be used to understand stock market behaviour and derive meaningful insights for decision-making.